

Stat 3113: Lesson 2

Introductory Material **Part 2**



Learning Objectives



Explore the nuances of observational studies and experiments.



Navigate through the intricacies of confounding and lurking variables.



Embark on a methodical journey with randomized comparative experiments.

Observation versus Experiment

- **OBSERVATION VERSUS EXPERIMENT**

Observe but don't disturb!

- An **observational study** observes individuals and measures variables of interest, but it **does not attempt to influence the responses**. The purpose of an observational study is to describe some group or situation.
- An **experiment**, on the other hand, deliberately imposes some **treatment on individuals to observe their responses**. The purpose of an experiment is to study **whether the treatment causes a change in the response**.
- When our goal is to understand **cause and effect**, experiments are the preferred source for fully convincing data.
- The **distinction between observational study and experiment** is one of the most important in statistics.





Example:

- Y = Lung cancer existence
- X = Number of cigarettes smoked
- How can this be done as an observational study with the sample being this class?
- How can this be done as an experiment with the sample being this class?

Example: Observational Study vs. Designed Experiments

Example : A study is conducted to determine the effect of cigarette smoking on the risk of lung cancer.

An Observational Study

- ✓ Find 100 people of which 50 have been smoking a pack a day for 10 years while the other 50 have been smoke free for 10 years.
- ✓ Measure lung capacity for each of the 100 people.
- ✓ Analyze, interpret, and draw conclusions from data.

Designed Experiment

- ✓ Find 100 people age 20 who do not currently smoke.
- ✓ Randomly assign 50 of them to the smoking treatment and the other 50 to the no smoking treatment.
- ✓ Those in the smoking group smoke a pack a day for 10 years while those in the control group remain smoke free for 10 years.
- ✓ Measure lung capacity for each of the 100 people.
- ✓ Analyze, interpret, and draw conclusions from data.

Notes:

- Observational studies can be our only source of data in some instances.

Ex: Study the bumper resistance of a car, based on the speed at which you're driving! (or windshield)

- If the goal is to understand cause and effect, experiments are the only fully convincing method of data collection.



A line graph with four data points connected by lines, showing an overall upward trend. The points are located at approximately (1, 2), (2, 4), (3, 3.5), and (4, 5).



Ex:

- 1 factor $\rightarrow$ Treatment = Level
- ≥ 2 factors $\rightarrow$ Treatment = Combination of levels

Ex:
#cigarettes + sleep time $\begin{cases} 1 \text{ pack} + 2 \text{ hrs of sleep} \\ 2 \text{ packs} + 8 \text{ hours of sleep} \end{cases}$

Example 1:

- A group of students *→ Cause and Effect!* wanted to determine what might influence plant growth. The students varied location of the plant (full sun, partial sun), water type (bottled, tap), and soil (top, potting).
- In order to study this, they used the same type of plant seed for all trials. Two of each possible combination was done in a random order.
- The height of the plant after 2 months was recorded.



Q1: What is an individual observation?

→ single data point
The height of a plant after 2 months under a specific experimental condition.

Q2: What is the response?

The height.

Q3: What is (are) the factor(s)?

- Location
- Soil
- Water type

Q4: What are the levels of each factor?

- Location (Full sun, partial sun)
- Soil (top, potting)
- Water type (bottled, tap)

Q5: List all possible treatments. $2 \times 2 \times 2$

Location	Water Type	Soil
Full Sun	Bottled	top
		potting
	tap	top
		potting
Partial Sun	Bottled	top
		potting
	tap	top
		potting

(2^3 Factorial Experiment)



Example 2:

- A group of students wanted to determine what might influence distance a dart is thrown from the center of the bullseye. The students varied distance the individual was from the board when throwing (3ft, 5ft, 7ft) and the type of dart (brand 1, brand 2, brand 3). After each throw, the distance the dart landed from the center of the bullseye is recorded.
- In order to study this, one person who was a good dart player was used for all throws. Each treatment was repeated five times and the order was randomized with a sufficient break between each throw.

Q1: What is an individual observation?

The distance a dart landed from the center of the bullseye under a specific throwing condition.

Q2: What is the response?

The distance from the center of the bullseye.

Q3: What is (are) the factor(s)?

Q4: What are the levels of each factor?

- ① Throwing distance from the board:
- 3, 5, 7 ft
- ② Type of dart
- Brand 1, 2, 3.

Q5: List all possible treatments.

Throwing distance	Type of dart
3 ft	Brand 1
	Brand 2
	Brand 3
5 ft	Brand 1
	Brand 2
	Brand 3
7 ft	Brand 1
	Brand 2
	Brand 3

(3² Factorial Experiment)

Example 3:

- Study of welfare system for single, working mothers. A job training program was offered to 5000 single, working mothers.
- Five years later their incomes were researched, and the results read, “the women who participated in the program were making significantly more money than those who did not participate.”

Q1: Is this an experiment or observational study? Why?

No. Because:

- Lack of random assignment
- Observational nature (The researcher didn't actively manipulate the participants' conditions)

Q2: What is the response variable?

Income of the single, working mothers.

Q3: What is (are) the factor(s)?

Participation in the job training program.

Q4: What are the levels of each factor?

- Participation
- Non-participation.

Q5: List all treatments.

1 factor => Treatment = Level.

Q6: Does this study show the job program caused the increase in income?
Explain.

No.! Correlation doesn't imply causation!

Because of other potential influences such as:

- Education background
- Prior work experience
- age
- Job opportunities in a specific industry.
- etc.

These variables are referred to as linking variables.

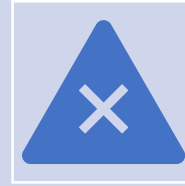
Lurking Variables and Confounding



A **lurking variable** is a variable that is not among the explanatory or response variables in a study but that may influence the response variable.

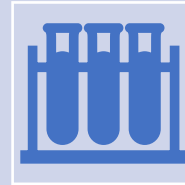


Observational studies of the effect of one variable on another often fail because of **confounding** between the explanatory variable and one or more lurking variables.



Two variables (explanatory variables or lurking variables) are **confounded** when their effects on a response variable cannot be distinguished from each other.

- Ex*
- New diet and Weight Loss (Exercise)
 - Education level and Income (Age $\rightarrow$ Work Experience)
 - Smoking and Lung Cancer (Exposure to pollutants)
 - Sleep time and productivity (level of stress)



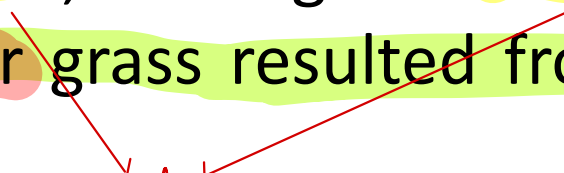
Well-designed experiments take steps to avoid **confounding**.

Confounding versus Lurking

❑ Two variables are **confounded** when you can't tell which of them (or whether it's the combination) had an effect on the response variable.

Example: You might want to test a fertilizer on your lawn. Suppose you spread it on half the lawn to see if the grass will look better there. If you spread it on the sunny half, leaving the shady half unfertilized, you won't know whether the greener grass resulted from fertilizer or sunshine (or the two together).

Sunshine

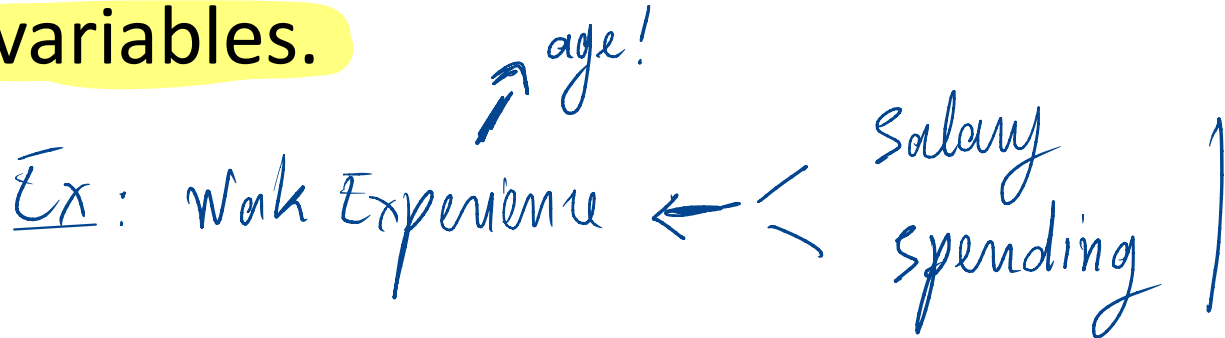


Because of the way the data was collected (bad experiment in this case), you cannot separate the effects of the fertilizer and the sun amount from each other.

Confounding versus Lurking

- ❑ A **lurking variable** is sometimes referred to as **common response**. It's a **variable that drives two other variables, creating the impression of a causal relationship** between them.

Example: There is a **strong positive correlation** between the **foot length** of K-12 students and **reading scores**. Clearly the **lurking variable is age**. Age is the **common link** between K-12 foot length and reading scores. It explains **what's happening with both variables**.



Randomized Comparative Experiments (1 of 2)

The remedy for confounding is to perform a comparative experiment in which some individuals receive one treatment and similar units receive another. Most well-designed experiments compare two or more treatments.

Ex: Sunshine!

Some experiments may include a control group that receives an inactive treatment or an existing baseline treatment.

In absence of treatment!

An experiment that uses both comparison of two or more treatments and random assignment of subjects to treatments is a randomized comparative experiment.

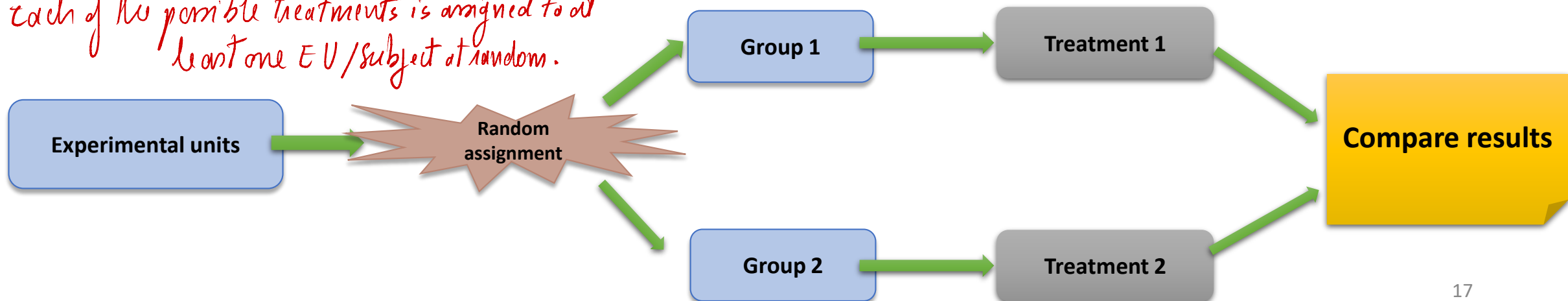
Randomized Comparative Experiments (2 of 2)

- Comparison alone isn't enough. If the treatments are given to groups that differ greatly, bias will result. The solution to the problem of bias is **random assignment**.
- In an experiment, **random assignment** means that individuals are assigned to treatments at random, that is, using some sort of chance process. *(Random number generator)*
It allows us to equalize the effects of unknown or uncontrollable sources of variation.

COMPLETELY RANDOMIZED DESIGN (CRD)

- In a **completely randomized experimental design**, all the subjects are allocated at random among all the treatments.

Each of the possible treatments is assigned to at least one E U / subject at random.



The logic of randomized comparative experiments

- **PRINCIPLES OF EXPERIMENTAL DESIGN**

- The **basic principles** of statistical design of experiments are:

- **Control** the effects of lurking variables on the response, most simply by comparing two or more treatments.

- Making conditions as similar as possible for all treatment groups

- **Randomize**—use chance to assign subjects to treatments.

- **Replicate**—use enough subjects in each group to reduce chance variation in the results.

NB: If the number of replicates is the same for each treatment combination $\Rightarrow$ Balanced!

- An observed effect so large that it would rarely occur by chance is called **statistically significant**.

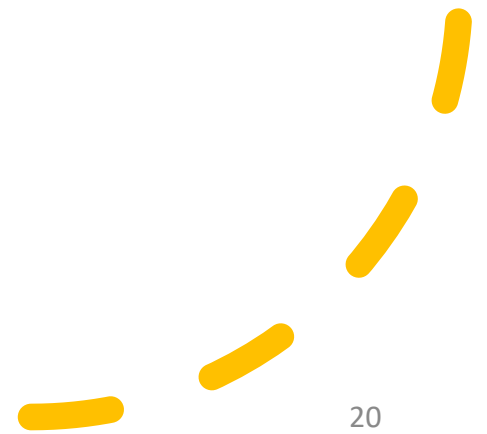
- A statistically significant **association** in data from a well-designed experiment **does imply causation**.

- Note: The logic of a randomized comparative experiment depends on our ability to treat all the subjects the same in every way, except for the actual treatments being compared.

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Reading

Chapter 1 – Section 9.1



Next Topic



Introductory Material Part 3



Chapter 9 - Section 9.5